

PROJECT TITAN

Design Decision Log

Educational Engineering Project

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Revision	A
Status	Released - Educational Baseline
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IMPORTANT SCOPE NOTICE

Project TITAN is an educational design and analysis project. It is not a clinically validated device, regulatory submission, or product cleared for human use.

Document Control and Revision History

Revision	Date	Description	Prepared By	Status
A	July 2026	Initial retrospective decision log documenting the baseline Project TITAN design and analysis choices.	Michael Lozada-Longog	Released

Source baseline: Project TITAN Engineering Design Report (July 2026), CAD assembly, completed engineering drawings, and Fusion 360 static FEA results documented by the author.

1. Purpose and Use

This log records why key engineering choices were made, which alternatives were considered, and what evidence supports each choice. Entries are retrospective where the original decision was made during CAD development before the formal documentation suite existed. Future entries should be created at the time of decision.

2. Decision Status Definitions

Status	Meaning
Released	Accepted for the current educational baseline
Conditional	Accepted subject to future verification or test
Superseded	Replaced by a later decision
Open	Decision required before the next build stage

3. Decision Register

Decision ID	Decision	Alternatives Considered	Rationale	Status	Evidence
DDL-001	Use a dual-side-plate architecture	Single-side hinge; four-bar linkage; powered joint	Dual plates provide a balanced, simple, and analyzable structure for the first mechanical baseline.	Released	PRD/SYS
DDL-002	Use a 10 mm nominal main-shaft diameter	Smaller pin; larger shaft	The 10 mm diameter aligned with the selected simplified bearing bore and provided a conservative starting geometry. Hand calculations remain required.	Conditional	CAD/CAL future
DDL-003	Use 22 mm OD × 10 mm ID × 7 mm wide simplified	Bushings; custom bearing; larger bearing	Standardized geometry simplified sourcing, CAD, and	Conditional	CAD/MFG future

Decision ID	Decision	Alternatives Considered	Rationale	Status	Evidence
	bearings		the load-transfer concept. Final COTS part number is not yet selected.		
DDL-004	Use 6 mm plate and frame thickness	5 mm; 8 mm; ribbed section	The 6 mm baseline balanced modeling simplicity and stiffness. Static FEA met the SF requirement; fatigue and physical tests remain open.	Released	FEA
DDL-005	Use 60 mm-wide side plates	Narrower tapered plate; wider plate	A 60 mm baseline provided room for mounting holes, bearing bore, and lightening slots while retaining edge distance.	Released	CAD
DDL-006	Reduce lightening-slot width after packaging review	Original 18 mm slot; no slots	The original 18 mm concept conflicted with the desired wall and hole edge distances. Narrower slots preserved material around load paths.	Released	CAD review
DDL-007	Place two strap slots at the top of the upper frame and bottom of the lower frame	Single central slot; no slots	Separated slots provide a clear conceptual attachment feature and distinguish upper/lower frame functions. Actual strap design is outside baseline scope.	Conditional	CAD/HF future
DDL-008	Use separate upper and lower frame components	Single continuous rail	Separate parts support modularity, independent drawing control, and future adjustability.	Released	PRD/CAD
DDL-009	Use two spacers	No spacers; integrated shoulders	Separate spacers simplified axial-stack control and allowed individual manufacturing documentation. A tolerance stack-up remains required.	Released	CAD
DDL-010	Use 6061-T6 aluminum for structural baseline	7075-T6; 316L; titanium; composite; PEEK	6061-T6 offered the best balance of cost, machinability,	Released	MAT-001

Decision ID	Decision	Alternatives Considered	Rationale	Status	Evidence
			availability, corrosion resistance, and adequate static performance.		
DDL-011	Model bearings as simplified solid components	Full rolling-element geometry; omit bearings	Simplified solids reduced model complexity while preserving envelope and load-transfer surfaces for educational FEA.	Released	CAD/FEA
DDL-012	Use component-level FEA before full-assembly FEA	Full assembly only	The isolated plate study validated material, constraints, load application, meshing, and solver workflow before contact-heavy assembly analysis.	Released	FEA
DDL-013	Apply full-assembly loads to inner bearing surfaces	Load side-plate bore directly	Inner-surface bearing loads better represented the intended external load entering through the bearing/shaft interface.	Released	FEA
DDL-014	Use a minimum safety-factor requirement of 2.0	1.5; 3.0	A target of 2.0 provided a clear educational static requirement. It is not a fatigue or clinical safety factor.	Released	PRD/FEA
DDL-015	Use refined mesh for final baseline results	Default mesh only	Refinement increased confidence that the displayed stress and displacement were not solely coarse-mesh artifacts.	Released	FEA
DDL-016	Accept refined-study results as the current static baseline	Earlier 100 N or preliminary assembly result	The refined study used corrected bearing load placement and current constraints. Reported values: 54.95 MPa, SF 2.377, 0.004 in displacement.	Released	FEA report
DDL-017	Create individual	Assembly drawing	Individual	Released	Drawing package

Decision ID	Decision	Alternatives Considered	Rationale	Status	Evidence
	part drawings	only	drawings demonstrate manufacturability, part control, dimensions, materials, tolerances, and revision identity.		
DDL-018	Assign sequential TITAN part numbers	File names only	Part numbers improve configuration control, BOM clarity, and drawing traceability.	Released	Drawing package
DDL-019	Treat bearings as COTS concept components	Custom-manufacture bearings	Commercial bearings reduce manufacturing scope and are appropriate for the intended architecture. Final supplier selection remains open.	Conditional	MFG future
DDL-020	Keep the project explicitly educational and non-clinical	Present as a finished medical device	The design lacks prototype, fatigue, human-factors, biocompatibility, and clinical evidence. Clear scope prevents unsupported claims.	Released	All documents
DDL-021	Defer GD&T until it can be applied correctly	Add decorative GD&T callouts	Incorrect GD&T would reduce credibility. Datum schemes and functional tolerances will be added after fit and inspection requirements are defined.	Released	MFG future
DDL-022	Defer powered actuation and sensors to V3	Add motor/electronics to baseline	A mechanical-first baseline isolates structural learning and avoids expanding scope before the hinge is physically validated.	Released	Roadmap
DDL-023	Revisit shaft material before physical V2	Retain aluminum without review	Bearing wear and shaft surface hardness may make stainless or	Open	MAT/RSK

Decision ID	Decision	Alternatives Considered	Rationale	Status	Evidence
			hardened steel preferable for a functional prototype.		
DDL-024	Require physical validation before wearable testing	Proceed directly to human testing	FEA alone cannot validate contacts, fatigue, wear, assembly error, or misuse. Benchtop testing is required first.	Released	RSK/PRO future

4. Key Decisions Requiring Revisit Before Prototype V2

Decision	Reason for Revisit	Required Evidence
Main shaft material and finish	Current aluminum baseline may be weak in wear and surface hardness	Bending calculation, bearing-interface review, material trade
Bearing retention and fits	Current assembly does not release a production retention method	Tolerance/fit study, supplier data, prototype pull-out test
Fastener selection	Current rigid joints are CAD abstractions	Fastener grade, size, torque, thread engagement, locking method
Side-plate geometry	Static pass does not establish fatigue or weight optimum	Fatigue analysis, topology study, physical strain/deflection data
Mechanical range of motion and hard stop	Current baseline lacks validated ROM limit	Kinematic requirement, collision review, hard-stop design and test

5. Design Decision Template for Future Entries

Required Entry Fields

Decision ID; date; problem statement; alternatives; criteria; selected option; rationale; supporting evidence; affected documents/parts; risks; verification action; approval/status.

6. Decision Quality Review

The baseline decisions show a coherent mechanical-first development strategy. The strongest decisions are those directly supported by completed evidence: modular CAD, component drawings, FEA workflow, refined static results, and material baseline. The weakest or most conditional decisions concern production fits, shaft material, fasteners, dynamic loading, fatigue, and human use because those areas do not yet have calculations or physical test evidence.

7. Conclusion

The design decision log converts the project history into a traceable engineering narrative. It demonstrates that Project TITAN evolved through packaging constraints, assembly troubleshooting, simulation setup correction,

and documentation improvements. Future revisions should record decisions prospectively and link each entry to drawings, calculations, risks, tests, and change requests.

References

Project TITAN Engineering Design Report, July 2026.

TITAN-PRD-001, Product Requirements Document, Revision B.

TITAN-SYS-001, System Architecture Document, Revision A.

TITAN-MAT-001, Material Trade Study, Revision A.

TITAN-RSK-001, Educational Risk Analysis, Revision A.